

Bangla E-Commerce Sentiment Analysis Optimization using Tokenization and TF-IDF

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Abstract—This paper aims to explore the sentiments of Bangla customer reviews and provide valuable insights for potential buyers. The proposed methodology involves a comprehensive approach to data collection from online shopping platforms, carefully removing duplicates and irrelevant information and labeling sentiments. Trigram features were used in combination with TF-IDF for effective feature selection. Data augmentation was also utilized to address the class imbalance. Various classifiers were deployed to categorize sentiments, including Naive Bayes, Random Forest, Extra Tree, KNN, Logistic Regression, Support Vector Machine, Artificial Neural Network, and Multilayer Perceptron. In analyzing Bangla product sentiments, the Multilayer Perceptron model performed exceptionally well and achieved an accuracy rate of 96.42% using the proposed methodology. The Artificial Neural Network closely followed it with 96.23% and Naive Bayes with 95.38%, providing valuable insights into the Bangla customer sentiment analysis landscape.

Index Terms—Natural Language Processing, Sentiment Analysis, Product Reviews, Bangla Natural Language Processing (BNLP), Machine learning, Online Shopping, TF-IDF, Bangla Text Analysis.

I. INTRODUCTION

E-commerce platforms have transformed how people shop in the digital age. With their convenient user interfaces, they have become essential for businesses to prioritize customer opinions and product reviews [1]. Customer opinions and product reviews have become invaluable tools in guiding informed purchase decisions, with over 95% of customers relying on online reviews before making a purchase [2]. In Bangladesh's growing e-commerce market, understanding customer sentiment is crucial to ensure trust and product quality [3]. By analyzing these sentiments, businesses can make informed decisions and maintain a positive online reputation,

ultimately benefiting customers in identifying top-performing products [4]. The Bangla language, with its rich cultural and historical significance, presents a challenge in the realm of Natural Language Processing (NLP) due to the need for computer-readable datasets and resources [5]. Despite past research in Bangla NLP, there is a need for more advanced tools and methodologies to advance the field [6]. Recent developments in NLP, driven by statistical methods, have transformed how we interact with computers and analyze sentiment in various media channels [7]. Sentiment analysis, a crucial aspect of NLP, empowers analysts to understand user opinions, fostering a more compelling online community [8]. This research paper focuses on conducting a sentiment analysis of Bangla product reviews. By combining innovative feature selection methods and diverse classification models, we aspire to contribute to the refinement and advancement of sentiment analysis techniques tailored specifically for the Bangla language.

The rest of the paper is organized as follows. Section 2 examines existing literature and methodologies on sentiment analysis in Bangla, with previous approaches highlighted. In Section 3, the data collection process and analysis are outlined. Section 4 presents the methodology used for sentiment analysis, including the criteria for feature selection and the implementation of classification models. Section 5 presents the analysis results, evaluating the techniques used and discussing their implications. Finally, in Section 6, the paper summarizes the findings and proposes potential avenues for future research in Bangla sentiment analysis.

II. LITERATURE REVIEW

Sentiment Analysis, or opinion mining, is an essential aspect of natural language processing (NLP) that uncovers

the feelings and viewpoints expressed through written or spoken words. In this section, we delve into the research conducted on sentiment analysis and language processing in the Bangla language. We also discuss the progress made in this field and the remaining obstacles.

Researchers from [9] developed a weighted lexicon data dictionary (LDD) for sentiment analysis on Bangla datasets. They applied techniques such as normalization, tokenization, and stemming to extract sentiments using LDD and BTSC algorithms. Additionally, they utilized supervised machine learning classifiers and achieved an accuracy rate of 82.21% by incorporating BiGram features. In another Research [10], researchers utilized deep-learning models to forecast sports-related comments in Bangla. They compared the results of CNN, LSTM, and BiLSTM algorithms. Among the three models, CNN using glove word embedding demonstrated the highest accuracy rate of 94.57%. In [11], Researchers emphasized the importance of sentiment analysis for better decision-making in the e-commerce industry. They used various algorithms, including NB, LR, SVM, RF, KNN, and DT, and compared their results with previous studies. Using a Support Vector Machine (SVM), they achieved a 94% accuracy rate for the Bangla dataset.

In [12], researchers proposed using the Term Frequency-Inverse Document Frequency (TF-IDF) technique with Trigram features to predict user emotions from Bangla text about products on e-commerce websites. They employed an SVM classifier with the highest accuracy of 90.68% for sentiment prediction. Another work, by the same authors [13], introduced a novel approach to sentiment analysis in the Bangla language using Deep Learning models. They implemented several models, including hierarchical attention-based LSTM, D-CAPSNET-Bi-LSTM, and BERT with LSTM, and achieved accuracy values of 78.52%, 80.82%, and 84.18%, respectively. In a different study [14], researchers presented a new approach for sentiment analysis of Bangla text reviews using a particular version of the Recurrent Neural Network called Long Short-Term Memory. Their proposed model achieved an accuracy of 84% and a precision of 85%. In [15], researchers suggested using machine learning models to search for hate speech in Bengali comments on Facebook. Their model categorizes seven types of hate speech by extracting and encoding data from the comments using attention-based LSTM, GRU decoders, and 1D convolutional layers. With an accuracy rate of 77%, hate speech in Bengali comments on social media has been successfully identified. In [16], researchers proposed a system that can classify customer reviews into positive and negative classes based on their sentimental feedback. Using multinomial Naive Bayes, their proposed system can classify restaurant reviews with 80.48% accuracy.

In [17], researchers introduced a highly efficient system that accurately detects sentiments in Bangla sentences. This system demonstrated an impressive accuracy rate of 90%. Researchers in [18] conducted research to identify offensive content written in Bengali across different websites. The study employed diverse deep learning and machine learning algorithms such as LSV, LR, NB, RF, ANN, and RNN. Among all the examined algorithms, the RNN utilizing deep learning achieved the highest accuracy of 82.20%. Researchers from [19] presented a cutting-edge technique for assessing the sentiment of Bengali comments. They express Bengali words as vectors using the word2vec approach to do this. They categorized Bengali comments with a 75.5% accuracy rate by fusing word2vec word co-occurrence and sentiment polarity scores.

As we explore sentiment analysis in the Bangla language, we have discovered that Researchers studying sentiment analysis in Bangla have used various methods to achieve accurate results. For example, one study used a weighted lexicon data dictionary and achieved 82.21% accuracy, while another used CNN and glove word embedding in a sports context and achieved 94.57% accuracy. In the realm of e-commerce sentiment analysis, SVM was used and attained 94% accuracy. Another study proposed TF-IDF with Trigram features and yielded 90.68% accuracy. Deep learning models achieved accuracy rates ranging from 78.52% to 84.18% in one study, while another focused on hate speech detection on Facebook and reached 77% accuracy. Several other studies explored various algorithms, with accuracy rates ranging from 75.5% to 90%, contributing to the evolving landscape of Bangla sentiment analysis. These diverse approaches are collectively helping to advance sentiment analysis in the Bangla language.

III. PROPOSED METHODOLOGY

This section will discuss the approach and methodology used for our Bangla product sentiment analysis research. Our methodology involves five steps: data collection, pre-processing, tokenization, data augmentation, and feature selection. We used the TF-IDF technique with trigram features and applied multiple classification models for feature selection. Figure 3 summarizes our approach.

A. Dataset Collection

The initial dataset was collected from various social media pages, groups, and online shopping platforms like Daraz and ChalDal.com, consisting of 16,311 customer reviews. However, the initial dataset contained entries in Bangla, English, and Banglish (Bangla written using English characters). As our research focused on Bangla sentiment analysis, we eliminated all non-Bangla entries to create a refined dataset with authentic Bangla comments and reviews. This resulted in a final dataset of 5,016 data

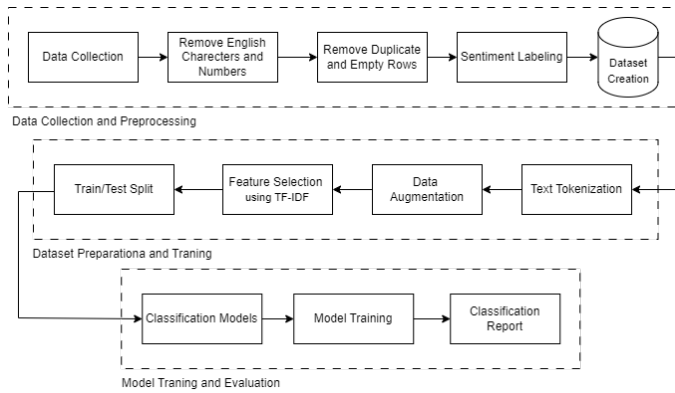


Fig.3: Proposed Methodology

rows. We then manually labeled each customer’s review as ‘positive’ or ‘negative’ based on the sentiment expressed in the review.

Table I: Sentiment Analysis Dataset

Comment	Sentiment
প্রোডাক্ট কোয়ালিটি অনেক ভালো। এটা যে স্টিজ তা জানা ছিল না। ঠিক ভাবে ইউজ করলে ৩/৪ বছর অনাহাসে চলে যাবে।	Positive
খুব বাজে, দামটাও অনেক বেশি।	Negative
খুব সুন্দর অনেক অনেক পছন্দ হইছে কেউ চাইলে নিতে পারেন।	Positive
আমি অভার করছি একটা দিয়েছে আরেকটা পুরো ফুলতু জিনিস আমি অভার করছিলাম এই পিক দেওয়া আছে ওটা আর আমার কাছে পাঠিয়েছে চাদ এর ওটা।	Negative
দাম অনুপাতে ভালোই।	Positive

Among these, 2,652 were identified as "positive," while 2,364 were identified as "negative." With positive evaluations accounting for approximately 52.9% of the total and negative evaluations constituting about 47.1%, the dataset demonstrates a well-balanced distribution of positive and negative sentiments. This balanced distribution comprehensively represents customer feedback and enables a robust analysis of both positive and negative sentiments [20].

B. Data Preprocessing

The data preprocessing stage is crucial in preparing the dataset for further research. Various techniques are used to prepare the dataset for further analysis during this stage. We prepare the dataset for further analysis by following a set of steps.

- **Removing Duplicate and Empty Rows:** We perform a duplicate check on the dataset to identify and elim-



Fig.3: Positive Sentiment Word Cloud



Fig.3: Negative Sentiment Word Cloud

inate duplicate rows, ensuring each review is unique and not repeated. Additionally, we remove any rows with missing or null values, as they do not contribute to the analysis.

- **Removing English Characters:** After the first step, we remove English characters from the reviews. This step is essential to ensure that the sentiment analysis is based solely on Bangla text and not influenced by English elements.
- **Removing Bangla Numbers:** In this step, we removed both Bangla and English numerical values from the dataset. This stage is crucial since sentiment analysis should concentrate on textual rather than numerical information.
- **Mapping Sentiment Labels:** Initially, the dataset's sentiment column has text labels like "positive" and "negative." In this final step, we convert these sentiment labels into binary values to simplify categorization. We give '0' for negative sentiments and '1' for positive sentiments. This modification makes the dataset more compatible with the machine learning algorithms.

C. Text Tokenization

We used the `word_tokenize()` function from the `nltk` package to tokenize the text data to simplify further analysis. The `word_tokenize()` function breaks down each

sentence in the 'comment' column of the dataset into individual words or tokens. This procedure transforms The raw text into a series of understandable syllables, making it more straightforward to process and evaluate [21]. The data frame's new column, 'tokenized_comment,' contains the tokenized words processed using tokenization. This column finds a set of tokens corresponding to each row's related comment.

D. Data Augmentation

To mitigate the class imbalance in the dataset, we adopted a comprehensive approach encompassing both oversampling and undersampling techniques [22]. Employing oversampling, we leveraged the RandomOverSampler() method to synthetically increase the representation of the minority 'negative' sentiment class by replicating existing samples, creating an oversampled dataset denoted as X_{over} and y_{over} . This successfully balanced the number of samples for both 'positive' and 'negative' sentiment classes. Simultaneously, we performed undersampling with the RandomUnderSampler(), which selectively removed samples from the 'positive' class until equilibrium was achieved with the 'negative' sentiment class, yielding a final augmented dataset denoted as $X_{\text{augmented}}$ and $y_{\text{augmented}}$, thereby effectively addressing the class imbalance issue.

E. Feature Selection using TF-IDF

: Gradient Boosting is a machine learning we used trigram features and the Term Frequency-Inverse Document Frequency (TF-IDF) vectorization method to choose the features. We divided the supplemental data into training and testing sets, X_{train} and X_{test} , and their associated sentiment labels, y_{train} , and y_{test} . This is crucial to guarantee an objective evaluation of the model. Trigram features offer more contextual information to the TF-IDF vectorizer, which improves the model's capacity to recognize sentiment patterns [24].

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \quad (1)$$

Here, $\text{TF}(t, d)$ represents the term frequency of term (t) in document (d), which is the number of times term (t) appears in document (d). $\text{IDF}(t)$ is the inverse document frequency of term (t), calculated as:

$$\text{IDF}(t) = \log \left(\frac{N}{1 + \text{DF}(t)} \right) \quad (2)$$

(N) is the total number of documents in the corpus, and ($\text{DF}(t)$) is the number of documents that contain the term (t). The training data (X_{train}) are then fitted and transformed using the TF-IDF vectorizer to provide numerical representations of the comments using trigram features ($X_{\text{train_tfidf}}$). After that, we train the sentiment classification models using this converted data. To guarantee that the model's performance is consistently assessed on unobserved data ($X_{\text{test_tfidf}}$),

the testing data (X_{test}) is converted using the fitted TF-IDF vectorizer [23].

F. Classification Models

Our research utilized various classification models to analyze patterns and make predictions. The Naive Bayes (NB) algorithm assumes feature independence when predicting class labels based on the Bayes theorem. Random Forest (RF) generates multiple decision trees during training and combines their predictions through voting. Extra Tree (ET), an extension of RF, employs a unique randomization strategy to enhance diversity in decision trees. The Artificial Neural Network (ANN) mimics the structure of the human brain, utilizing interconnected neurons in input, hidden, and output layers. Multilayer Perceptron (MLP), a type of neural network, addresses regression and classification tasks with densely interconnected neurons. K-Nearest Neighbors (KNN) predicts class labels based on most of the K nearest neighbors in the feature space, using the Euclidean distance for proximity determination. Logistic Regression uses the logistic sigmoid function to establish relationships between input features and binary outcomes. Finally, the Support Vector Machine (SVM) is a robust algorithm for classification, finding optimal hyperplanes in high-dimensional feature spaces to separate different classes [25].

IV. RESULTS AND ANALYSIS

After conducting tests on all of the machine learning algorithms mentioned above, using the dataset, we analyzed the performance of each algorithm by evaluating the Precision, Recall, F1 score, and Accuracy metrics. We implemented a strict experimental design to evaluate the test results to ensure a fair comparison. Precision is a crucial metric in binary classification tasks like sentiment analysis. It measures the accuracy of positive predictions by considering the proportion of true positives to the sum of true positives and false positives. Recall evaluates the model's ability to correctly identify true positive cases relative to the total positive cases in the dataset. The F1 score is a comprehensive metric combining precision and recall to provide a balanced assessment, particularly useful in imbalanced datasets. It is calculated as the harmonic mean of precision and recall. Additionally, accuracy is a fundamental metric that measures the overall correctness of model predictions by considering the ratio of correct predictions to the total predictions. A higher accuracy score indicates more accurate predictions, while a lower score suggests potential performance issues.

Table II provides an overview of the performance metrics achieved by various models using the proposed methodology for sentiment analysis on Bangla text. Among the models, Artificial Neural Network (ANN) and Multilayer Perceptron (MLP) emerged as top performers, demonstrating exceptional precision, recall, F1 score, and

Table II: Classification Report of Different Models

Models	Precision	Recall	F1 Score	Accuracy
NB	95.33	95.4	95.38	95.38
RF	93.62	93.53	93.57	93.59
KNN	92.24	91.95	93.54	92.08
ET	94.5	94.54	94.52	94.53
LR	94.33	94.42	94.34	94.34
SVM	94.8	94.86	94.81	94.82
ANN	96.2	96.29	96.22	96.23
MLP	96.39	96.47	96.42	96.42

accuracy. The ANN model achieved 96.20% precision, 96.29% recall, 96.22% F1 score, and 96.23% accuracy. Conversely, the MLP model outperformed with 96.39% precision, 96.47% recall, 96.42% F1 score, and 96.42% accuracy. On the other hand, the Naive Bayes (NB) model demonstrated the least favorable performance, with precision, recall, F1 score, and accuracy values of 95.33%, 95.40%, 95.38%, and 95.38%, respectively, it fell behind the ANN and MLP models in terms of overall accuracy. Therefore, the proposed methodology has effectively identified models with varying degrees of effectiveness in handling sentiment analysis on Bangla text, with ANN and MLP emerging as the most robust performers.

We conducted a parallel analysis to compare our proposed methodology with an alternative approach where TF-IDF and tokenization were not applied. Notably, without TF-IDF and tokenization, precision values exhibited a significant decrease for the models. Figure 4 demonstrates the precision values derived from various classification models. The MLP and ANN models emerge as the top performers with precision scores of 96.39% and 96.47%, respectively. However, precision values notably decrease when a regular methodology is employed without the proposed approach, with MLP achieving 80.22% and ANN achieving 79.58%. Figure 5 further illustrates the

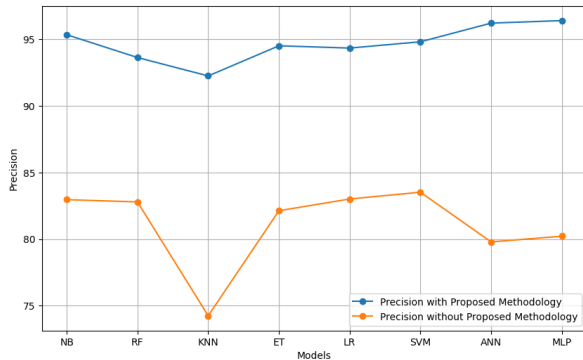


Fig.3: Proposed vs Regular Methodology (Precision)

impact of the proposed methodology on recall values. While MLP and ANN models demonstrate consistent excellence with recall scores of 96.29% and 79.19%,

respectively, these values notably drop without the proposed enhancements, with MLP and ANN achieving recall values of 79.58% and 79.66%, respectively. The methodology's crucial role in elevating recall metrics is thus reaffirmed. Figure 6 presents F1-Score values,

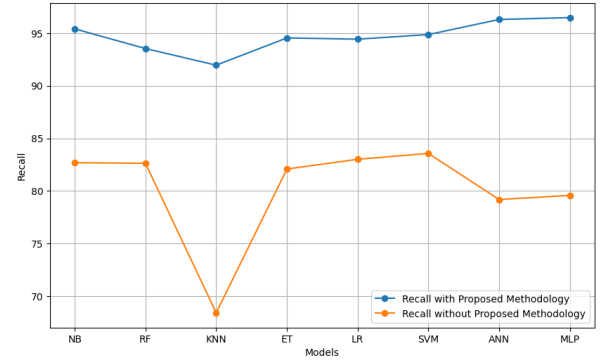


Fig.3: Proposed vs Regular Methodology (Recall)

highlighting the superiority of the MLP and ANN models. MLP attains an F1 score of 96.42%, while ANN achieves 96.34%. Conversely, a regular methodology devoid of the proposed enhancements yields lower F1 scores, with MLP at 79.66% and ANN at 79.86%. These findings underscore the profound impact of the proposed methodology in augmenting the overall model performance, particularly in terms of F1-Score metrics. Figure 7 illustrates the

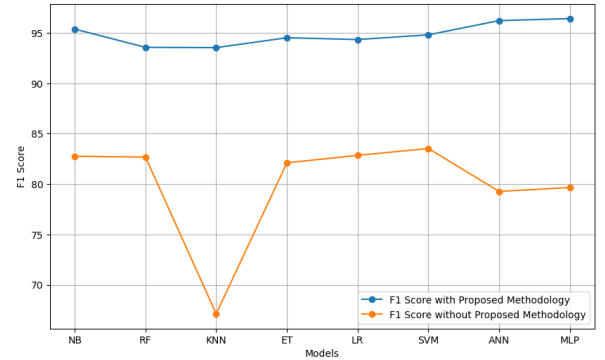


Fig.3: Proposed vs Regular Methodology (F1-Score)

accuracy of various classification models, with MLP and ANN again emerging as the leaders, achieving impressive accuracies of 96.42% and 96.23%, respectively. Conversely, traditional methodologies yield diminished accuracy rates, with MLP at 79.86% and ANN at 79.46%. This marked difference underscores the effectiveness of the proposed methodology in enhancing model accuracy. In conclusion, our research indicates that incorporating the proposed methodology significantly improves the performance of classification models in Bangla sentiment analysis. This underscores the critical importance of this approach in elevating model outcomes and its potential impact in practical applications.

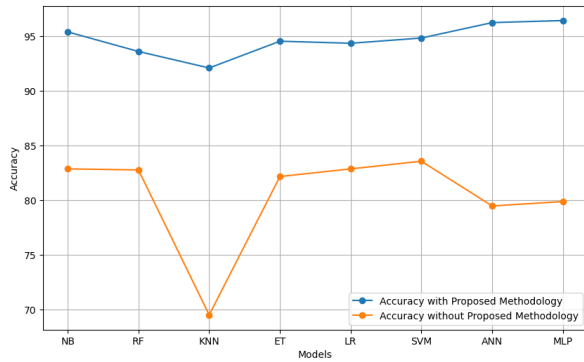


Fig.3: Proposed vs Regular Methodology (Accuracy)

V. CONCLUSION

Our research on Bangla Product Sentiment Analysis holds significant importance for NLP and sentiment analysis, given the complexity of the Bangla language and its unique challenges. We gained valuable insights into sentiment patterns prevalent in Bangla product reviews through a comprehensive study encompassing data collection, preprocessing, feature selection, and model exploration. We achieved highly precise predictions across multiple models by employing TF-IDF alongside trigram features to capture linguistic nuances. Notably, the proposed methodology yielded impressive accuracy rates, with the MLP model reaching 96.22%, while the KNN model achieved the lowest accuracy rate of 92.08% among those employing our methodology. Our findings bear practical implications for e-commerce and product review platforms, where an in-depth understanding of customer sentiment is vital for informed decision-making processes. Moving forward, future work could refine the proposed methodology to handle specific challenges inherent to Bangla language sentiment analysis, such as dialectical variations and context-dependent expressions, further advancing the Bangla Natural Language Processing field.

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